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A Mathematical Method for the Investigation of Inter-Element Effects in X-Ray Fluorescent Analyses

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Several investigators have tackled the problem of inter-element effects in X-ray fluorescent analyses from the theoretical aspect,[1] whilst others have taken a more empirical approach and interpreted their practical results in terms of calibration curves and graphs.[2] This paper suggests, from the broadest assumptions, what form these empirical formulæ are likely to take, and investigates how far these simple linear extrapolations can be used in practice.

X-RAY fluorescent analysis provides a means of determining accurately and rapidly the amounts of the various elements present in an alloy, and is particularly suitable for concentrations ranging from 0·1% to 100%. A complicating factor in the use of this method is the modification of fluorescent emission caused by inter-element effects. The intensity of the radiation from one element may be reduced by partial absorption by other elements in the matrix or, alternatively, its fluorescence may be enhanced by their presence. These effects are systematic and to a large extent predictable, and in the following pages a mathematic method for investigating the phenomena is presented.

Theoretical Considerations

Let us imagine a binary alloy containing two elements A and B, and let us imagine further that the fluorescent radiation of A does not excite B, and also that, for all wavelengths, the absorption coefficients of the two elements are very similar. In these circumstances, admittedly unlikely in practice, the calibration curve of A (intensity of A radiation $v.$ percentage of A present) will be a straight line, as shown in Fig. 1.

If, in the case of binary alloy AC, the element C strongly absorbs the A fluorescent radiation, a calibration curve of the type shown in Fig. 2 will result. The straight line calibration curve for A in alloy AB is also included in Fig. 2, the intensities for 0% and 100% A being identical in the two cases.

It will be seen from Fig. 2 that, in the case of ternary alloy ABC, containing, say, 60% A, the intensity of the A radiation can vary from I_1 to I_2, depending on the relative concentrations of B and C in the remaining 40%.

Assumptions

At this stage it is proposed to make the following sweeping assumptions: the practical results will indicate their worth :—

(1) Apart from absorption effects, mutual enhance-

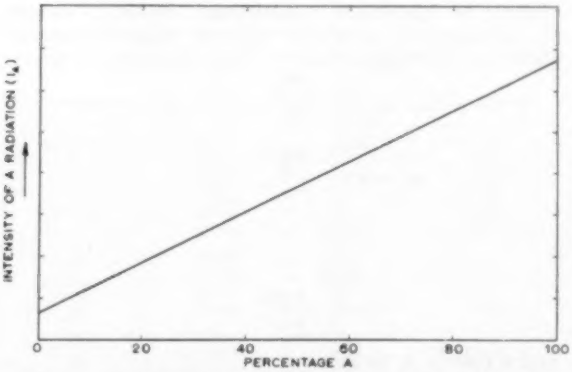

Fig. 1.—Calibration curve for element A in a binary alloy AB in which B does not absorb the fluorescent radiation of A.

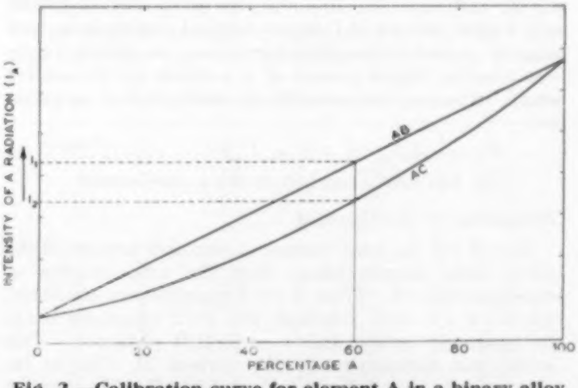

Fig. 2.—Calibration curve for element A in a binary alloy AC in which C absorbs the fluorescent radiation of A. The calibration curve for AB is repeated here.

ment also may occur. This will be regarded as negative absorption and assumed to obey the same laws.

(2) The absorption of element A by a percentage of C (say $P_C\%$) is linearly proportional to P_C. Thus, the actual departure, in terms of intensity, from the perfect AB binary calibration curves will be proportional to $P_C \times I_A$.

The second assumption is not unreasonable : over small ranges, twice as much C might well be expected to have twice as great an effect on the A radiation. Also, the proportion of A radiation absorbed by a given amount of C is likely to stay approximately constant regardless of the amount of A radiation. Thus, for different I_A's for a given P_C, the absorption is proportional to I_A.

The Form of the Equation

It is now possible to write down a projected inter-element effect formula. First, it is important to define the notation. All the percentages will be written as P with a subscript immediately after it referring to the element—as in P_C. At a later stage, when different samples are considered, another subscript will be added denoting the sample numbers. Thus P_{nm} refers to the percentage of the nth element in the mth sample. Similarly I_{nm} refers to the intensity of nth element in the mth sample. Using this notation, and taking the previous example,

$$P_{Am} = a_A + \beta_A I_{Am} + \gamma_A I_{Am} P_{Cm}$$

In practice we must consider more than one interfering element, and none of them is likely to be as similar to A as B was originally. In other words, to express the percentage of the nth element, in an alloy containing x elements, the formula which has been suggested so far is :—

$$P_{nm} = a_n + \beta_n I_{nm}(1 + \Sigma_x \gamma_{nx} P_{xm})$$

This would be very inconvenient to use, as the percentages of the other elements appear in the right-hand side of the equation and they are, of course, related by similar equations, and hence solution of such equations would be involved.

The equation could be transformed into an expansion of this type :—

$$P_{nm}=a_n+\beta_n I_{nm}\ (\lambda+\Sigma_x\mu_x I_{xm}+\Sigma_x\nu_x I^2_{xm}+\ldots\ldots\ldots\ldots)$$

where the λ, μ and ν coefficients are related to the constants a_n, β_n and γ_{nx}. As, however, we purposely neglected any higher powers of I_x in our original assumptions, and also as ν and succeeding terms can be shown to be composed of higher powers of γ_x's which are themselves small, it is not unreasonable to write a final equation as :—

$$P_{nm}=a_n+I_{nm}\ (\kappa_o+\Sigma_x\kappa_{nm}I_{xm})\ldots\ldots\ldots\ldots(1)$$

(β_n has been absorbed in the κ coefficients)

Evaluation of the Constant

Now if x is the total number of elements present in the alloy under consideration, then the total number of constants is $x+2$. Thus, if $x+2$ samples were available, whatever I's were obtained, the $x+2$ equations could be rigorously solved and a perfect fit obtained. This would not necessarily be the correct fit, due to inaccuracies in measuring the I's and the P's. In practice a much larger number of samples is used, and a least-squares fit adopted. This can be very conveniently worked out on a computer as follows.

Define $P^{Form.}_{nm}$ as the P_{nm} calculated from the formula, and $P^{Chem.}_{nm}$ as the chemical figure given on analyses from the sample. Consider the value $\triangle$ where

$$\triangle_{nm}=\frac{P^{Chem.}_{nm}-P^{Form.}_{nm}}{I_{nm}}$$

This is a measure of the percentage departure of the percentage, as determined on the spectrometer, from the actual analytical figure. It is important that $\triangle$ should be expressed in this way, for the following reasons. Imagine two samples, one with $P_n=5\%$ and the other with $P_n=30\%$. If we obtain X-ray percentages of $5{\cdot}3\%$ and $30{\cdot}3\%$, respectively, we have in each case got an absolute error of $0{\cdot}3\%$. However, the fractional error is worse in the 5% case, than in the 30% case. A better fit could be obtained if the 5% correction were improved, even at the slight expense of the 30% figure. To get the best fit possible, it is important to choose a_n, κ_0 and $\kappa_1, \kappa_2 \ldots \kappa_x$, so that $\Sigma_m \triangle^2_{nm}$ is minimised. Thus, as

$$\triangle_{nm}=\frac{P^{Chem}_{nm}-P^{Form}_{nm}}{I_{nm}}$$

we can write

$$\triangle_{nm}=\frac{P^{Chem}_{nm}}{I_{nm}}\qquad\frac{a_n}{I_{nm}}\ -\ \kappa_0\ -\Sigma_x\kappa_{nx}I_{xm}$$

Now define a matrix J_n as follows :—

$$J_n = \begin{array}{c|c} & x+2 \text{ columns} \\ \hline \\ m \text{ rows} & \begin{array}{ccccc} \frac{1}{I_{n1}} & 1 & I_{11} & I_{21}\ldots\ldots\ldots\ldots & I_{x1} \\ \frac{1}{I_{n2}} & 1 & I_{12} & I_{22}\ldots\ldots\ldots\ldots & I_{x2} \\ \vdots & & & & \\ \frac{1}{I_{nm}} & 1 & I_{1m} & I_{2m}\ldots\ldots\ldots\ldots & I_{xm} \end{array} \end{array}$$

a matrix P_n as follows :—

$$P_n= \begin{array}{c|c} & 1 \text{ column} \\ \hline \\ m \text{ rows} & \begin{array}{c} \frac{P^{Chem}_{n1}}{I_{n1}} \\ \frac{P^{Chem}_{n2}}{I_{n2}} \\ \vdots \\ \frac{P^{Chem}_{nm}}{I_{nm}} \end{array} \end{array}$$

and a matrix κ_n as follows :—

$$\begin{array}{cc} x+2 \text{ columns} & \\ \hline a_n,\ \kappa_0,\ \kappa_1,\ \kappa_2\ \ldots\ldots\ldots\ldots\kappa_x & 1 \text{ row} \end{array}$$

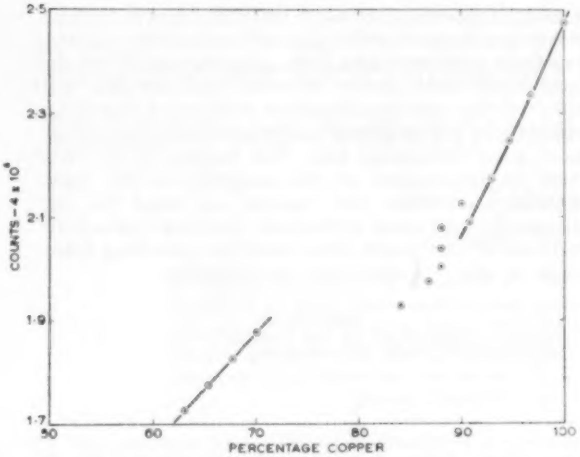

Fig. 3.—Graph showing percentage of copper plotted against counts. No corrections made.

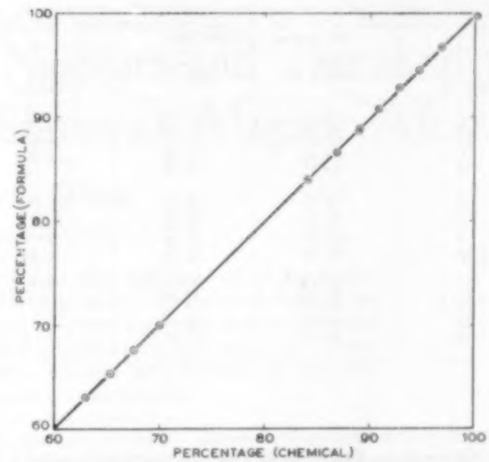

Fig. 4.—Graph showing the percentage of copper determined chemically plotted against the percentage obtained by application of the formula to the X-ray fluorescent results.

In this notation formula (1) can be written :

$$P_n = J_n \times \kappa_n \quad \ldots\ldots\ldots\ldots \quad (2)$$

It can be shown that $\Sigma_m \triangle^2_{nm}$ is minimised if κ is chosen to fulfil the following equation :

$$J^T_n \; P^{Chem}_n = J^T_n \; J_n \kappa_n \ldots\ldots\ldots\ldots \quad (3)$$

The procedure for evaluation is as follows. A large number of good samples are taken and their respective I_{nm}'s measured on a spectrometer. To work out the inter-element effect for the element " n," the equivalent J_n matrix is tabulated; similarly the P_n matrix is also tabulated. In the authors' case, these were then sent to the Ferranti Computer Centre at 21 Portland Place, London, W.1, which performed the necessary calculation and found the least-squares-fit the κ_n matrix. P^{Form}_{nm} can be calculated for each sample and the results examined to see how good a fit is obtained.

Practical Results

It was decided, before applying this method to practical problems, that a problem with an unnaturally wide range of percentages would be undertaken to test the various assumptions. With this aim in mind a series of samples was prepared by the British Non-ferrous Metals Research Association, consisting of straight brasses, straight bronzes and some gun-metals. It would be extremely unlikely that, in practice, one would ever wish to fit these widely different alloys on the same calibration curve, but their accurate chemical analyses were readily available. Also, it was known that tin had a profound absorption on copper, and that tin radiation could further increase the copper fluorescent radiation by mutual enhancement.

The chemical figures for the alloys are listed in Table I, and the X-ray intensities, obtained on a Solartron automatic vacuum X-ray spectrometer, Type XZ,1030, are listed in Table II.

The copper intensity/percentage graph is plotted in Fig. 3 to show the serious nature of the inter-element effect. It is interesting to notice that in the worst case a 7% shift in copper percentage is required to bring these points on to one straight line. The following equation was given as the solution to the inter-element effect on copper, after the results had been computed in accordance with equation (1).

$$P^{Form}_{Cu\,m} = 0 \cdot 78 + 10^{-4} I_{Cu\,m} (27 \cdot 98 + 4 \cdot 84 \times 10^{-4} I_{Cu\,m} - 0 \cdot 64 \times 10^{-5} I_{Zn\,m} + 1 \cdot 32 \times 10^{-4} I_{Sn\,m})$$

Table III and Fig. 4 give a comparison of the closeness of the fit.

The root mean square difference is 0·13%. The reproducibility on the spectrometer for the copper

TABLE I.—CHEMICAL ANALYSES OF COPPER ALLOY SAMPLES

Sample No.	Copper	Zinc	Tin
1	96·62	0	3·34
2	94·54	0	5·42
3	92·62	0	7·35
4	90·61	0	9·36
5	65·27	34·73	0
6	62·95	37·05	0
7	67·61	32·39	0
8	69·98	30·02	0
9	100·0	0	0
10	87·88	5·01	7·11
11	87·93	2·96	9·11
12	87·93	0·83	11·24
13	89·85	3·06	7·09
14	86·73	1·90	11·28
15	84·00	4·99	11·01

TABLE II.—X-RAY FLUORESCENT INTENSITIES OF COPPER ALLOYS

Sample No.	I_{Cu}	I_{Zn}	I_{Sn}
1	23,323	334	14,325
2	22,426	334	22,470
3	21,667	313	30,490
4	20,870	295	38,330
5	17,740	60,795	790
6	17,247	53,618	775
7	18,244	47,799	785
8	18,750	44,680	840
9	24,724	349	800
10	20,771	7,778	29,030
11	20,376	4,552	37,175
12	20,031	1,650	45,015
13	21,220	4,845	28,515
14	19,738	3,028	45,215
15	19,261	7,151	44,965

TABLE III.—COMPARISON OF COPPER CONTENTS DETERMINED BY CHEMICAL ANALYSIS AND BY X-RAY FLUORESCENT ANALYSIS SUITABLY CORRECTED

Sample No.	$P^{Chem}_{Cu\ m}$	$P^{Form}_{Cu\ m}$	Difference
1	96·62	96·78	+0·16
2	94·54	94·53	−0·01
3	92·62	92·85	+0·23
4	90·61	90·82	+0·21
5	65·27	65·27	0
6	62·95	63·03	+0·08
7	67·61	67·58	−0·03
8	69·98	69·94	−0·04
9	100·00	99·82	−0·18
10	87·88	87·64	−0·24
11	87·93	87·83	−0·10
12	87·93	87·94	+0·01
13	89·85	89·88	+0·03
14	86·73	86·60	−0·13
15	84·00	83·97	−0·03

intensity determination was 0·08% (standard deviation). It was considered that the chemical copper determination had been performed to a high accuracy and should not appreciably contribute to the error. It is worth noting, however, that although the three alloy types might give individually self-consistent chemical results, they might have some inter-group bias. The balance of the error must be apportioned to the simplicity of the inter-element corrections and, bearing in mind the unnecessarily large range attempted, the errors are satisfyingly small, and would, when used for correcting differences in any one alloy type, be negligible.

REFERENCES

1 Sherman, J., *Spectrochemica Acta*, 1955, **7**, 286–306.
2 Bareham and Fox, *J. Inst. Met.*, 1960, **88**, 344.

Stanton Balance for International Bureau

Stanton Instruments, Ltd., London, well-known manufacturers of precision, analytical, micro-chemical and thermo-recording balances, have recently supplied and installed a 10 kg. capacity high precision balance at the Bureau International des Poids & Mesures, Sevres, near Paris. The Bureau, which is the world's standards laboratory for those countries which adhere to the International Metric Convention, is re-equipping its Mass Section during the next few years. The 10 kg. balance is the first new high precision balance to be installed since 1910, and orders for further high precision balances have already been placed with Stanton Instruments, Ltd., for delivery during the next two years. The 10 kg. balance mentioned above is of the conventional three-knife edge design and is fitted with a special optical reading system. It will be used for the testing of larger weights and is expected to provide a precision of weighing of not less than 1 in 100 million.

Roto-Finishes Approved by A.I.D.

Roto-Finish, Ltd., have received A.I.D. approval for their Cahill electroless nickel process and their Tridur AL chromate conversion process for aluminium under D.T.D. Specification Nos. 900/4728 and 900/4729, respectively.

The Cahill electroless nickel process is a method of plating by chemical reduction and produces a semi-bright, hard (V.P.N. 450/550) coating of nickel/phosphorus alloy (90–92% nickel —8/10% phosphorus) on a variety of base metals; including steel, copper alloys, aluminium. By a suitable activation process it can be used for plating ceramics and phenolic mouldings.

The process is operated hot, 88/93° C., on a small scale in a glass or vitreous enamelled beaker, or on a larger scale in special glass lined tanks with electrothermal heating jackets. It requires no electric current or anodes; hence there is no problem in throwing power, and intricate and irregular shapes can be plated uniformly (±10% max. of the coating thickness). By a suitable heat treatment (1 hour at 400° C.) the hardness can be increased to 1,000 V.P.N. The process can be used for building up components machined undersize and for corrosion resistance, where it can replace expensive stainless or clad alloys. The exceptional hardness of the alloy plate makes it particularly suitable for use on components where wear is a problem.

Tridur AL conversion treatment for aluminium and its alloys gives a high rate of corrosion resistance and good paint adhesion. Coatings are produced in 20 seconds to a maximum of 3 minutes, in a yellow to khaki colour which can be dyed with various coloured aniline dyes if required.

Gas Analysis Licence Agreement

It is announced that Elliott Brothers (London), Ltd., a member of the Elliott-Automation group, has negotiated with the Office National d'Etudes et de Recherches Aeronautiques (O.N.E.R.A.) of France a licence to manufacture and sell two gas analysis instruments of advanced design. The analysers are of the paramagnetic type, a principle used in both thermo-pneumatic and thermo-magnetic convection systems of measurement. The territory covered by the licence is, in general, world wide with some exceptions of which the principal are France and former French territories, the Benelux countries, North and South America and Japan. The new instruments are primarily intended for the determination of oxygen content in gaseous mixtures and have already found extensive application in the control of furnace atmospheres; catalyst protection, flue gas analysis and the control of atmospheric test rooms are further examples of their use.

British Furnaces, Ltd., Derby Road, Chesterfield, have recently received orders for gas carburising equipment to be installed in the new Bathgate factory of the British Motor Corporation. These orders cover six Super Allcase sealed quench furnaces, complete with fully automatic sequence control; one continuous pusher-tray type gas carburising furnace, together with the necessary "RX" endothermic atmosphere generating units; and various other items of auxiliary equipment.

Schloemann A.G. have been awarded a contract by the Hüttenwerk Oberhausen A.G. for the construction of a continuous rod mill, complete with auxiliaries. It is to be laid out to roll $\frac{13}{64}$ in. to $\frac{21}{64}$ in. diameter rod from $3\frac{1}{8}$ in. square billets. The maximum delivery speed when producing $\frac{13}{64}$ in. rod will be 6,900 ft./min., and the rod mill will have a monthly output of about 30,000 tons.